GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

Similar Triangles; Indirect Measurement

Question Bank - Additional 6.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 and 2, decide whether the triangles are similar. Show the three ratios.

1. Sides 6, 8, 9 and 15, 20, 22.5

Similar — every ratio is 2.5

15/6 = 20/8 = 22.5/9 = 2.5.

2. Sides 4, 5, 7 and 6, 8, 10.5

Not similar — 6/4 = 1.5, but 8/5 = 1.6

The middle ratio differs.

For questions 3 through 5, the two triangles are similar. Find the missing sides.

3. Sides 5, 6 and 8; the copy's 5-side is 15.

18 and 24

k = 3.

4. Sides 20, 30 and 35; the copy's 20-side is 4.

6 and 7

k = 0.2.

5. Sides 2, 5 and 6; the copy's 6-side is 21.

7 and 17.5

k = 3.5.

For questions 6 and 7, the triangles are similar.

6. A 1.8-m woman's shadow is 1.2 m long. At the same moment a building's shadow is 16 m. How tall is the building?

24 m

Each height is 1.8 ÷ 1.2 = 1.5 times its shadow: 16 × 1.5.

7. Mia's eyes are 5 ft up, 2 ft from a mirror on the ground. In it she sees the top of a building 30 ft beyond the mirror. How tall is the building?

75 ft

k = 15; 5 × 15.

Choose one.

8. Which triangle is similar to one with sides 3, 5 and 6?

- ● **A. 9, 15, 18**
- ○ B. 5, 7, 8
- ○ C. 6, 10, 14
- ○ D. 3, 5, 9

A

Every side × 3.

9. A 6-ft man casts a 5-ft shadow. At the same moment a tree's shadow is 35 ft. How tall is the tree?

- ○ A. 29.2 ft
- ○ B. 36 ft
- ○ C. 210 ft
- ● **D. 42 ft**

D

$k = 35 \div 5 = 7$; 6×7.